Rubik's Cube Engine — Abu Bakr Structure (Optimized)

This document describes a Rubik's Cube simulation system based on a face-index structure. The model represents each face as a list of 9 elements and encodes each piece as an integer combining piece identity and orientation.

Core Idea: The cube is represented using faces rather than global cubie lists. Each position holds a piece ID and orientation packed into a single integer. This allows both visual clarity and computational efficiency.

Features: - Face-based representation - Integer encoding for performance - Orientation handling for edges and corners - Physically valid state verification - Rotation system based on adjacency mapping

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# Example core function (rotation snippet)

def rotation(t, face, sens=1):
    f = t[face]
    if sens == 1:
        f[0],f[2],f[4],f[6] = f[6],f[0],f[2],f[4]
        f[1],f[3],f[5],f[7] = f[7],f[1],f[3],f[5]
    else:
        f[0],f[2],f[4],f[6] = f[2],f[4],f[6],f[0]
        f[1],f[3],f[5],f[7] = f[3],f[5],f[7],f[1]
```